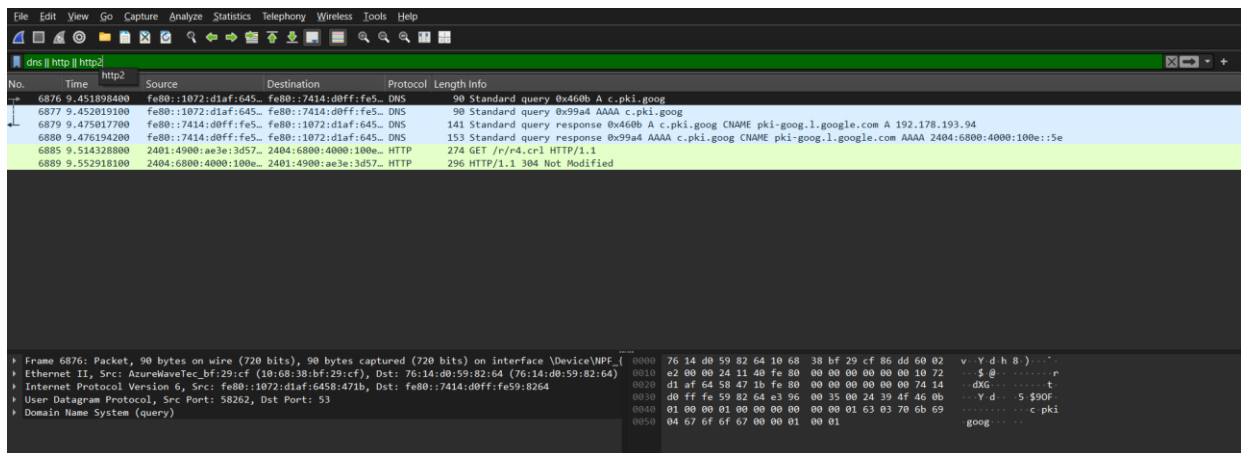


TASK-1



TASK-2

Run the following command in **Command Prompt** (Windows) or **Terminal** (Linux/macOS):

```
nslookup www.flipkart.com
```

Sample Output

Server: dns.google

Address: 8.8.8.8

Non-authoritative answer:

Name: www.flipkart.com

Address: 163.53.78.128

Observation

Website	IP Address Returned
---------	---------------------

www.flipkart.com	163.53.78.128 (example; your result may differ)
--	---

Brief Explanation

The nslookup command queries a **DNS (Domain Name System)** server to translate the domain name www.flipkart.com into its corresponding **IP address**. The returned IP address is the numerical address that your computer uses to locate and communicate with Flipkart's web server. The exact IP address may vary depending on your location, DNS server, and Flipkart's content delivery network (CDN).

TASK-3

Difference Between HTTP and HTTPS

HTTP

HTTP (HyperText Transfer Protocol) sends data in plain text and is **not** encrypted. It is **less secure** and can be intercepted by attackers. Uses **Port 80** by default.

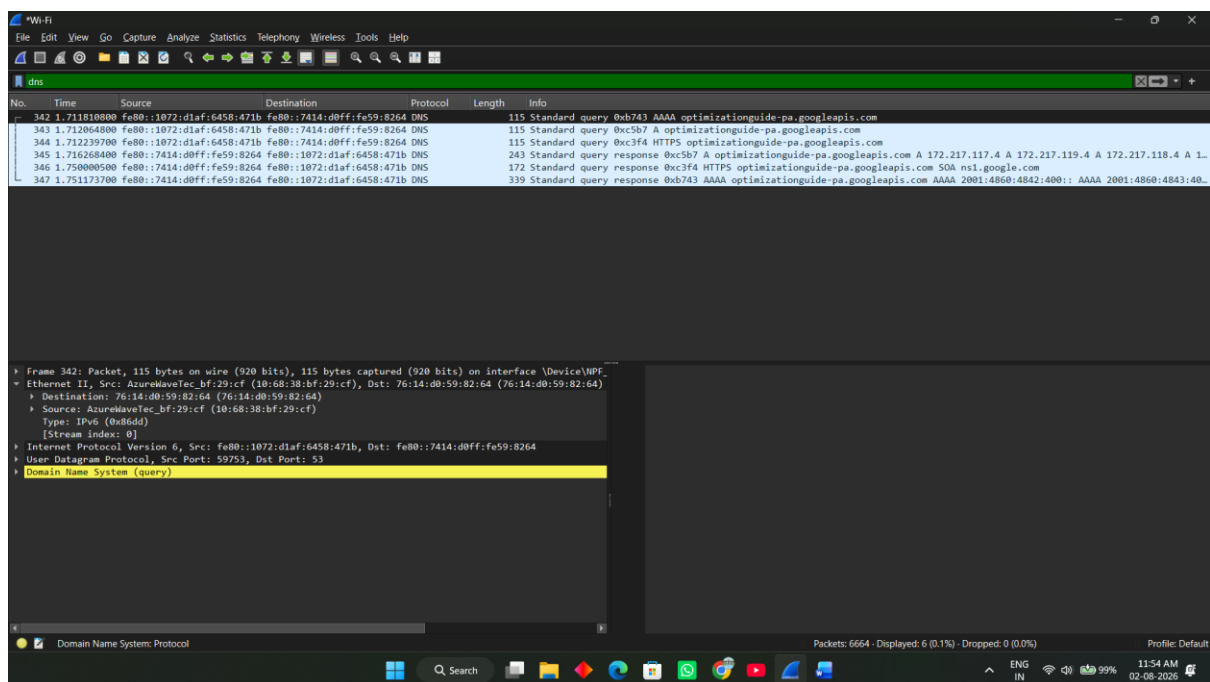
HTTPS

HTTPS (HyperText Transfer Protocol Secure) sends data in plain text and is **not** encrypted. It is **more secure** and protects data such as passwords, banking details, and personal information. Uses **Port 443** by default.

Real-World Example

A website that relies on **HTTPS** is **Spotify** (<https://www.spotify.com>). HTTPS encrypts the communication between your browser and Spotify's servers, protecting your login credentials, account information, and streaming activity from being intercepted by others on the network.

TASK-4



Here is the **short summary** of your Wireshark capture:

1. **Queries Sent (Client → Server):** Your machine requested **IPv6 (AAAA)**, **IPv4 (A)**, and **HTTPS** records for **optimizationguide-pa.googleapis.com** almost simultaneously.

2. **Responses Received (Server → Client):** The DNS server replied to each query ID with matching IPv4 addresses (172.217.117.4, etc.) and IPv6 addresses (2001:4860:...).
3. **Protocol:** Standard DNS running over **UDP (Port 53)** using IPv6 link-local addresses.

TASK-5

Protocol Chosen: DHCP (Dynamic Host Configuration Protocol)

Real-World Scenario:

When I visit a cafe and connect my phone to the free Wi-Fi to open **Instagram**, the **DHCP** protocol automatically assigns my phone an IP address, subnet mask, default gateway, and DNS server. This happens without any manual network configuration. Once the IP address is assigned, my phone can access the internet and load Instagram to view posts, reels, and messages.

Conclusion

DHCP makes connecting to a network quick and automatic by assigning the required network settings, allowing users to access online services like Instagram without manually entering IP configuration details.